



POWERED WEARABLE TECHNOLOGIES FOR DEMENTIA CARE: EVALUATING ACTIVITY RECOGNITION MODELS AND DATASET CHALLENGES

Keywords: Wearable Technology, Activity Recognition, AI, Elderly, Dementia

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Introduction to Dementia and Care Challenges

Understanding the Impact and Need for Monitoring

Rising Dementia Cases

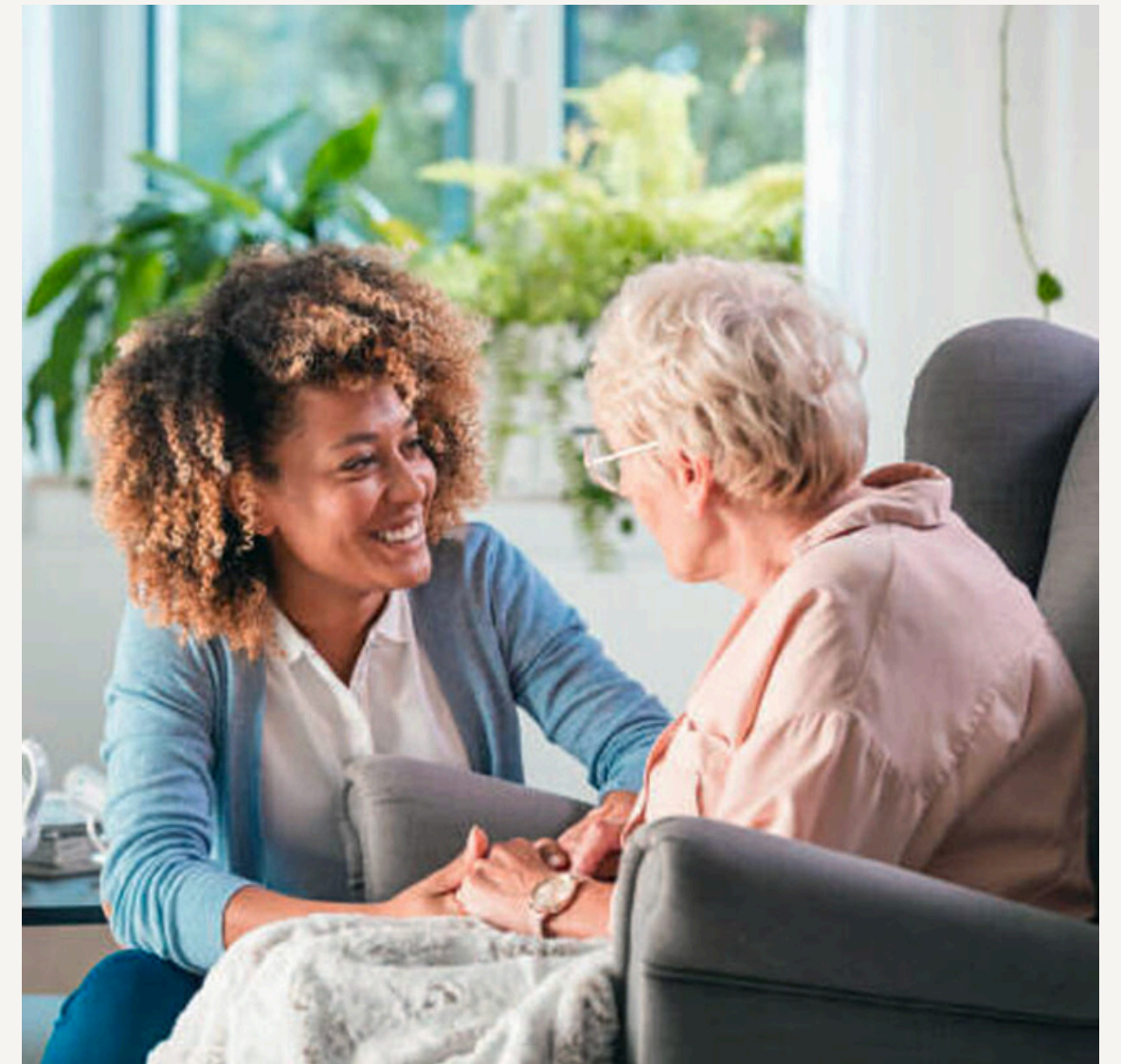
Over 55 million people worldwide suffer from dementia, expected to double by 2050

Caregiver Burden

High stress, depression, and physical strain reported among caregivers

Importance of Monitoring

Tracking movement, sleep, and health indicators improves safety and independence



The Role of Wearable Technology

Enhancing Dementia Care with AI-Powered Devices



CONTINUOUS MONITORING

Wearables track movement, sleep, and vital signs in real time



ANOMALY DETECTION

AI-powered wearables identify abnormal behaviors and alert caregivers

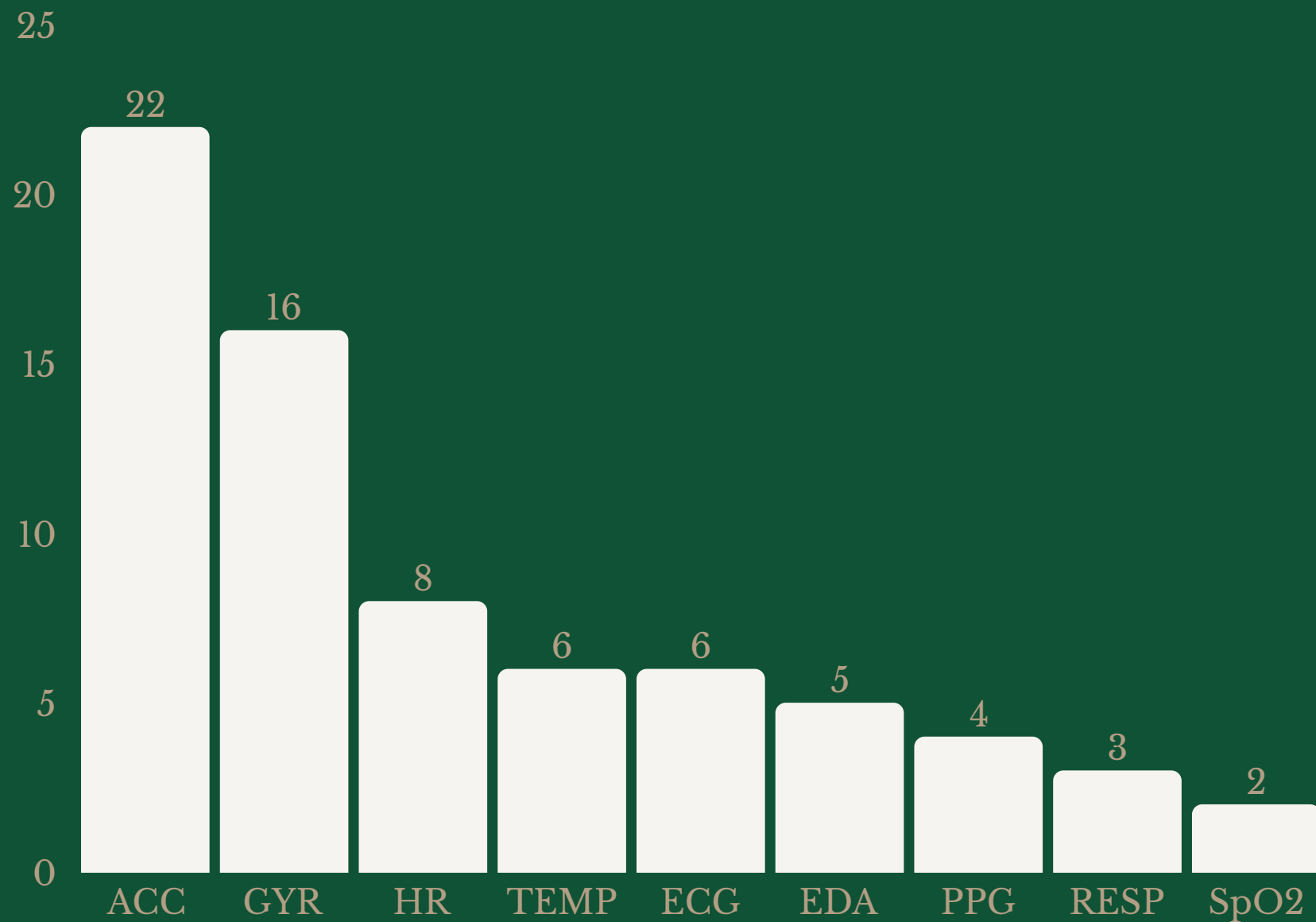


IMPROVED QUALITY OF LIFE

Supports independence and reduces caregiver burden through early intervention

Datasets for Activity Recognition

30 DATASETS ANALYZED



Sensors



Accelerometers and Gyroscopes
Track movement and orientation



Heart Rate and Physiological Sensors
Monitor cardiovascular activity and stress levels



Bio-Sensors
Electrodermal activity, temperature, and oxygen saturation provide additional health insights

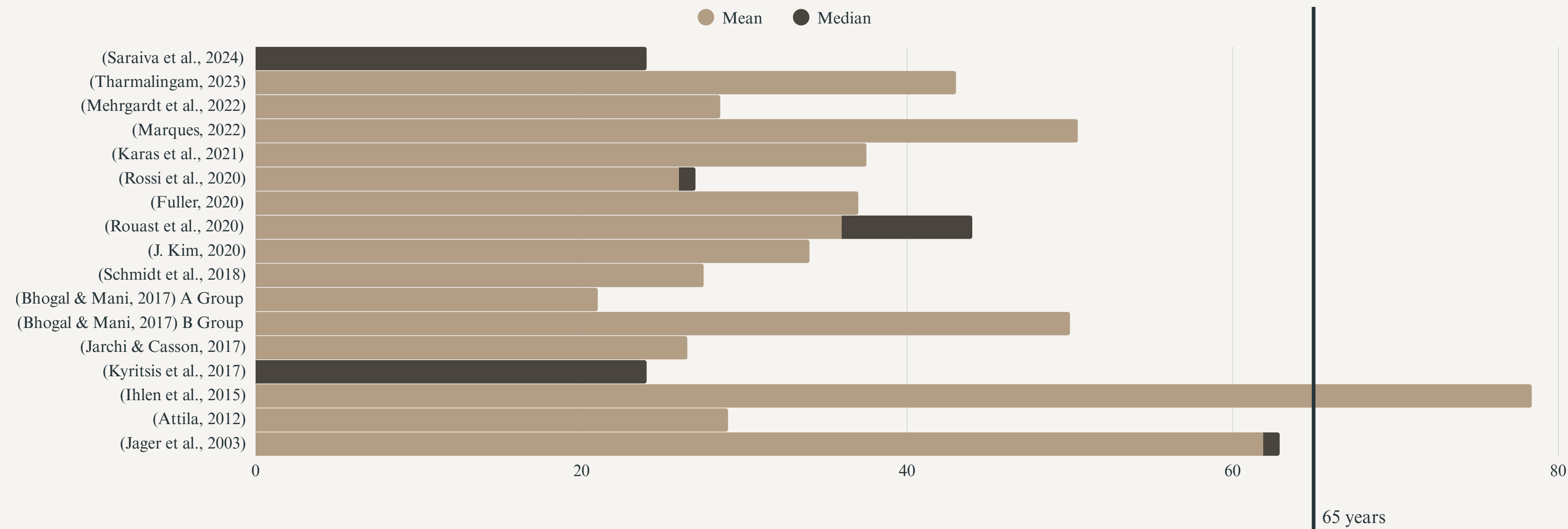
Participants Demographics

Underrepresentation of Elderly

Most datasets focus on younger adults, limiting model generalizability

Gender Imbalance

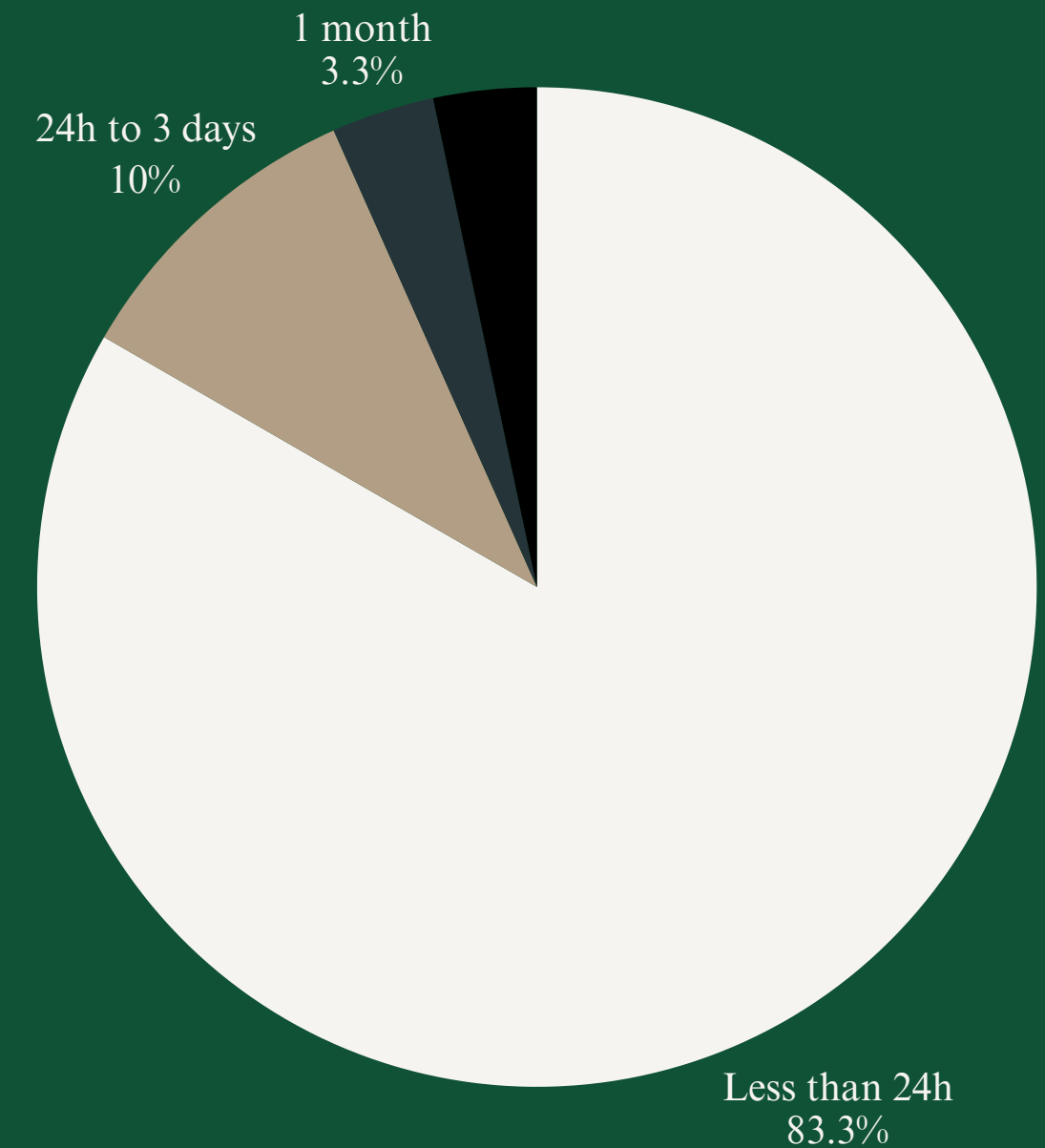
Some datasets lack gender diversity, leading to biased AI models



Duration of the Recordings

Most datasets capture less than 24 hours, reducing long-term analysis.

Long-duration data is essential for identifying behavioral changes



Commonly Tracked Activities

Walking (40%), Eating (20%), Sitting and sleeping (13.3% each)

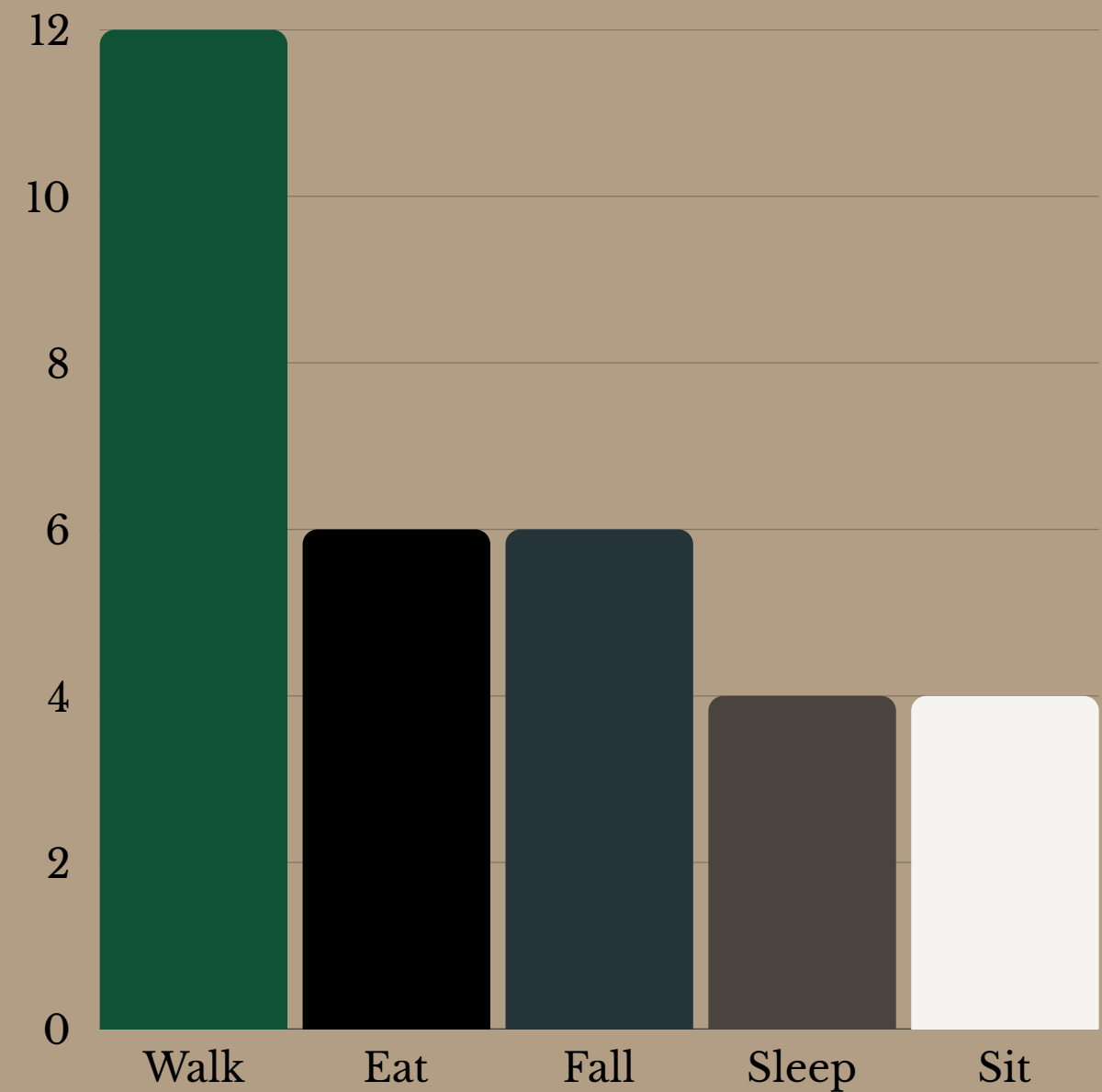
Specialized Activity Labels

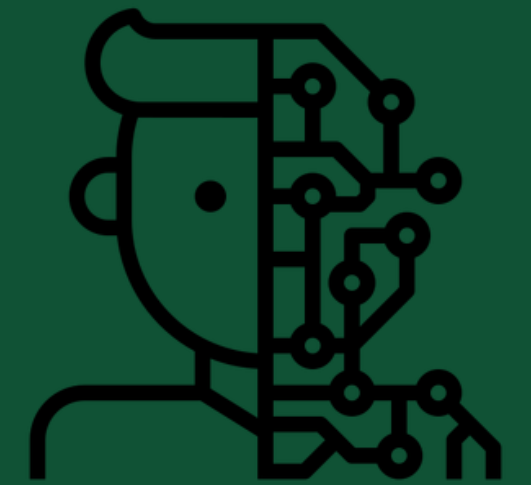
Includes Hygiene, wandering, fall detection, and motor skill tasks

Dataset-Specific Insights

Certain datasets track transitions, household activities, and stress levels

Recorded Activities





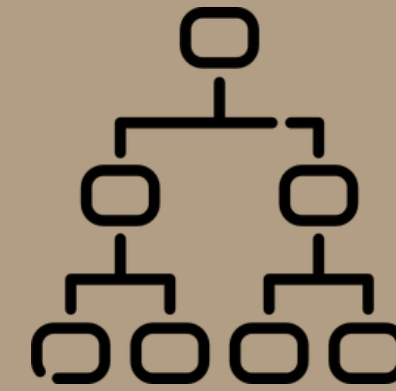
Artificial Intelligence

Simulation of human intelligence by machines



Random Forest

Tree based AI models

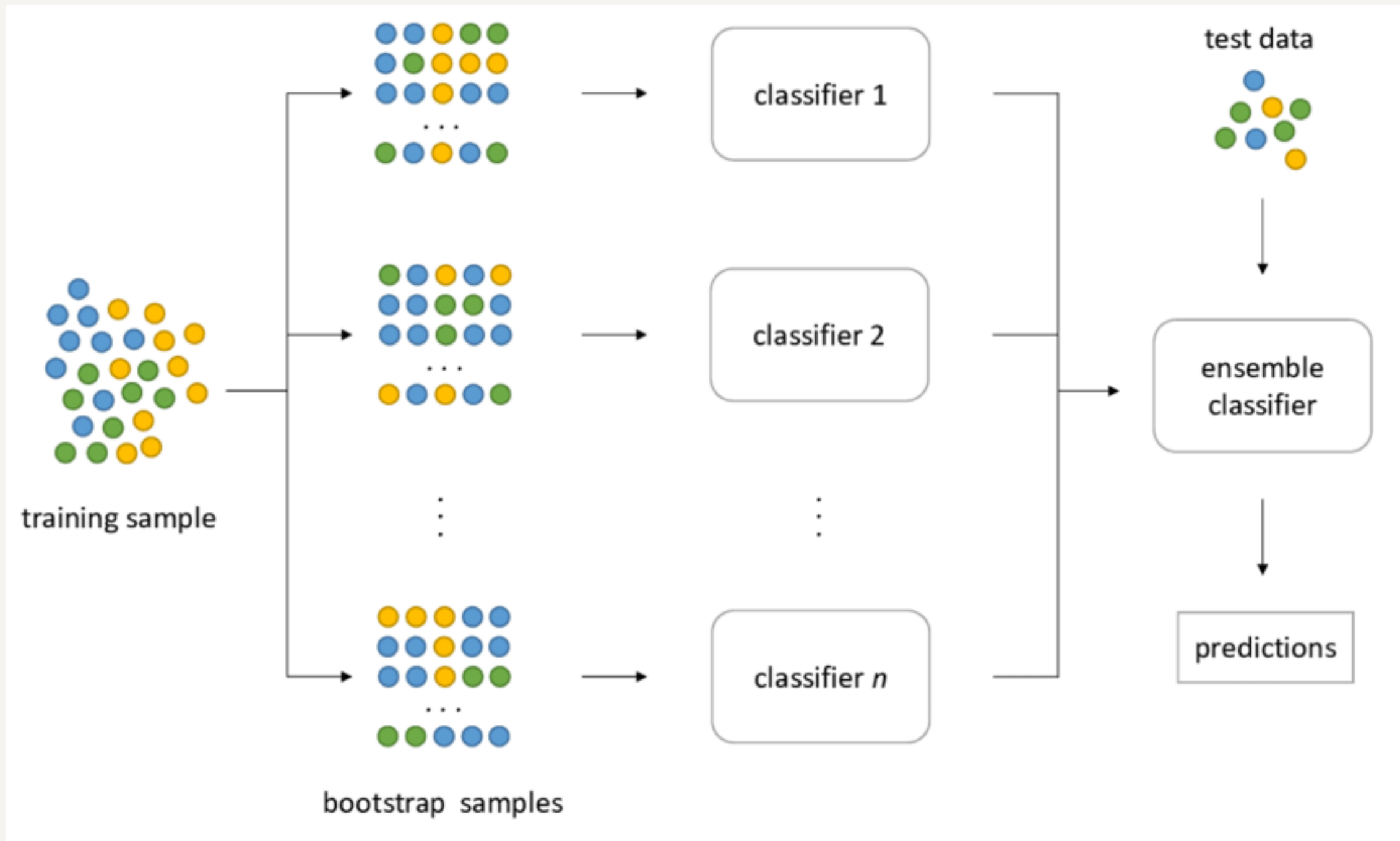


Bagging ensemble method

Independent Trees

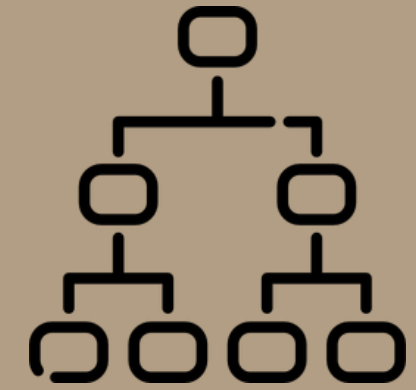
Grows multiple deep trees independently and averages their outputs

Parallel training of trees (faster on small data)



Extreme gradient Boosting (XGboost)

Tree based AI models

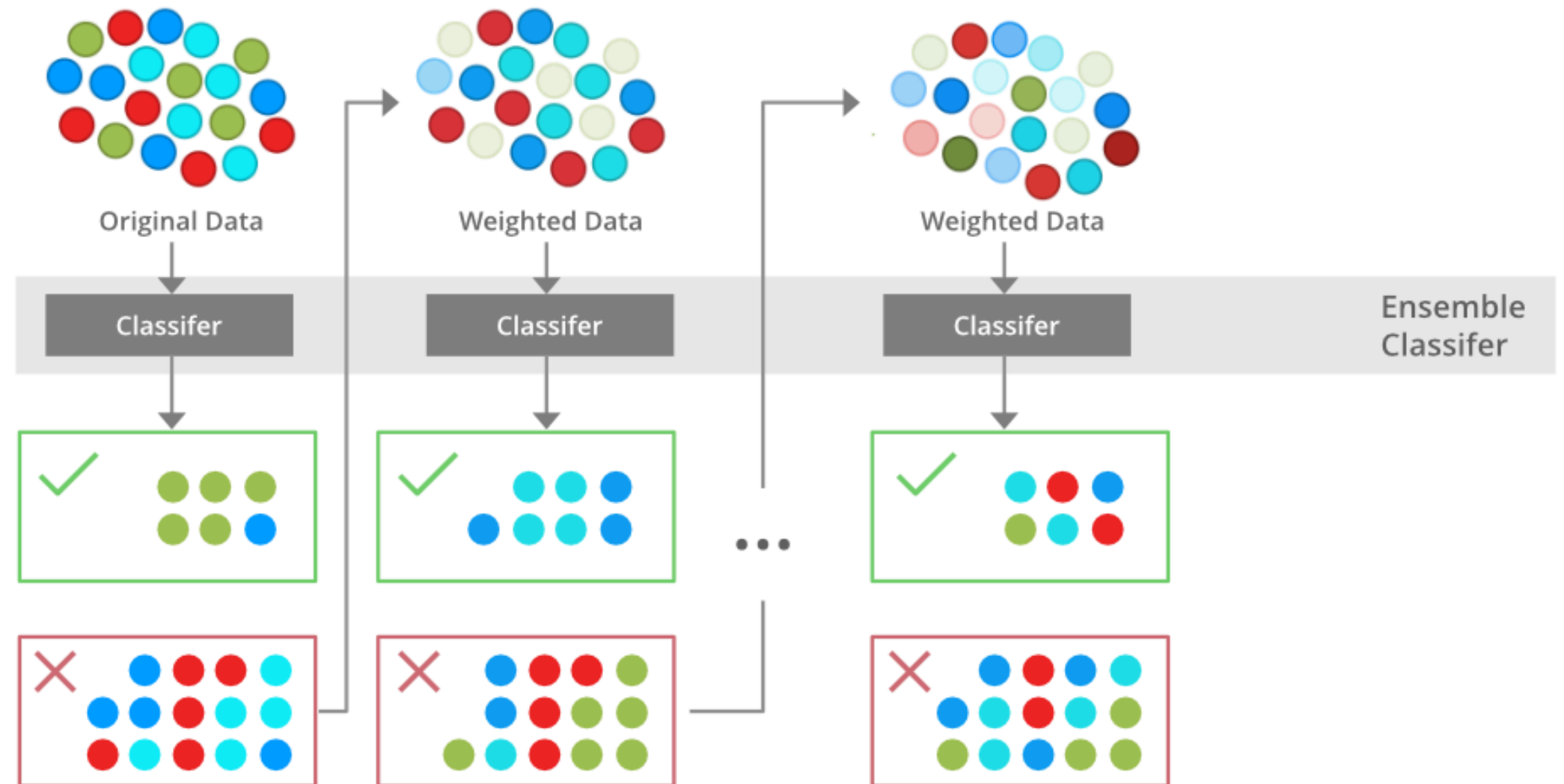


Boosting ensemble method

Sequential Trees (each tree corrects previous errors)

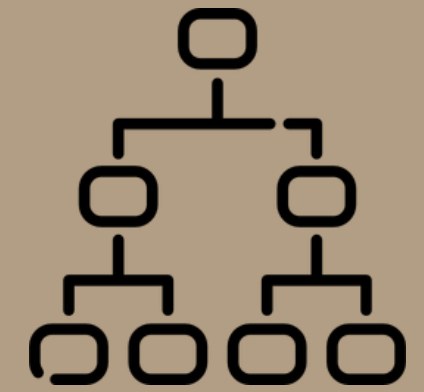
Grows trees sequentially, each one correcting the mistakes of the previous tree

Trees are built sequentially (can be slower, but optimized for speed)



LightGBM

Tree based AI models



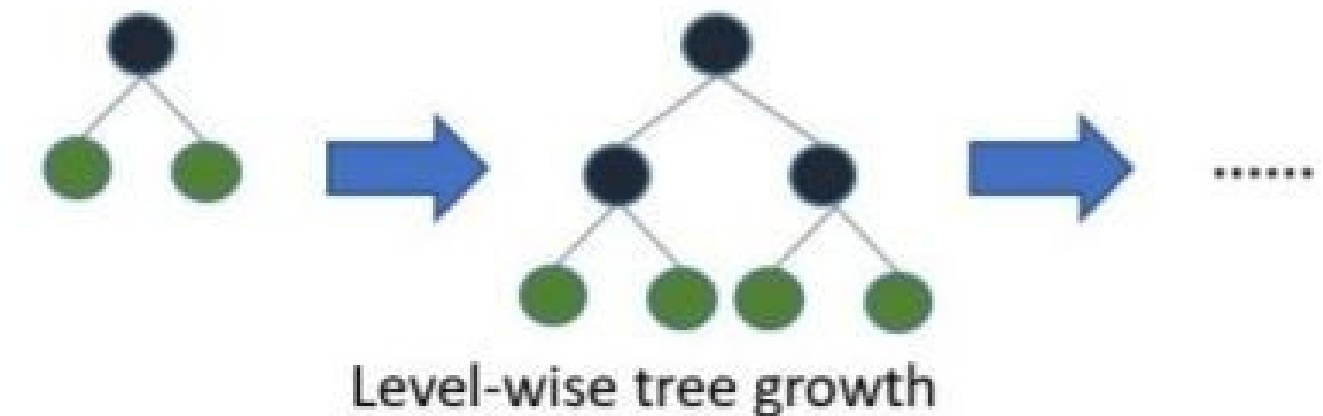
Boosting ensemble method

Similar to XGBoost, but uses histogram-based learning for efficiency

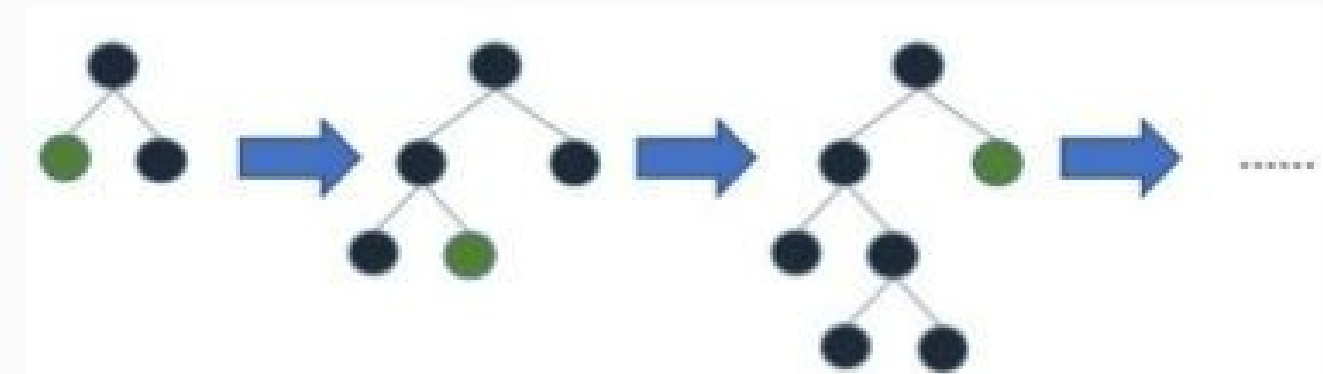
Faster than XGBoost, handles categorical data natively

Can be sensitive to hyperparameters, may overfit on small data

XGBoost:



LightGBM:

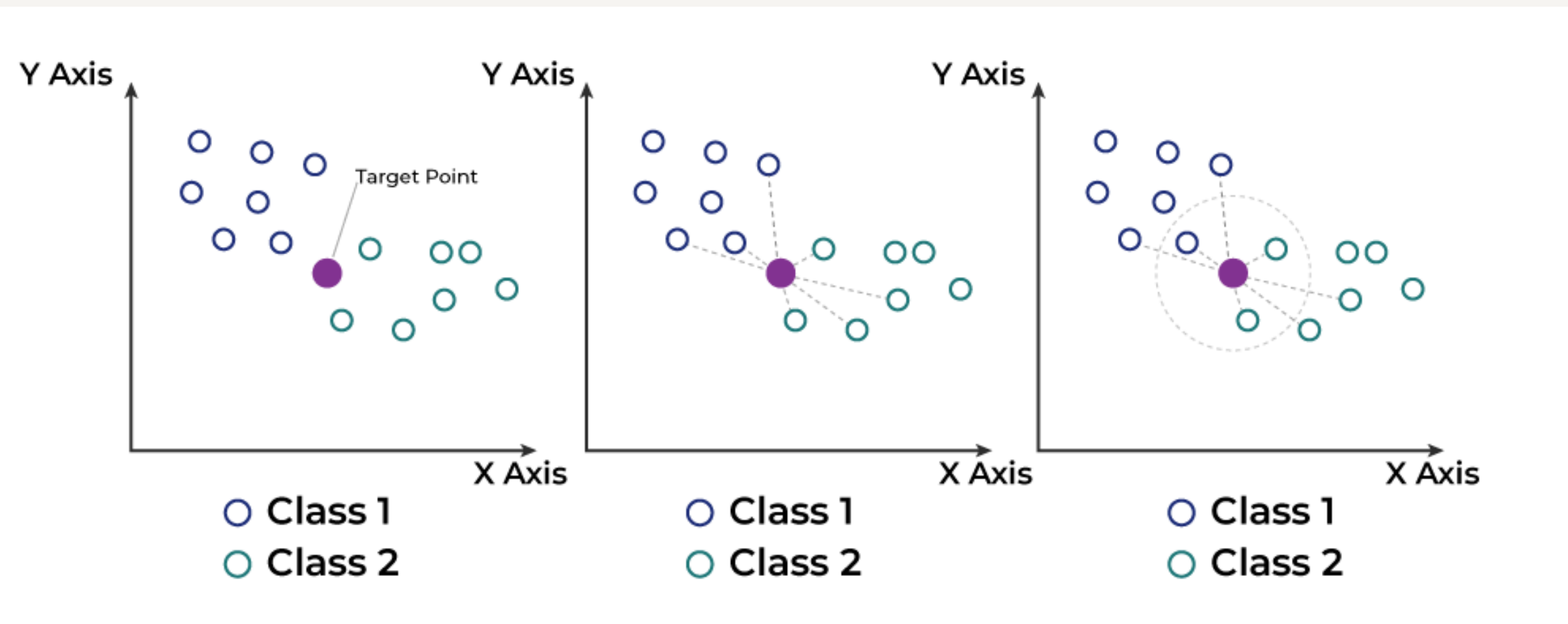


k-Nearest Neighbors (KNN)

AI models

Instance-based learning

Stores all training examples,
predicts based on majority
class of nearest neighbors



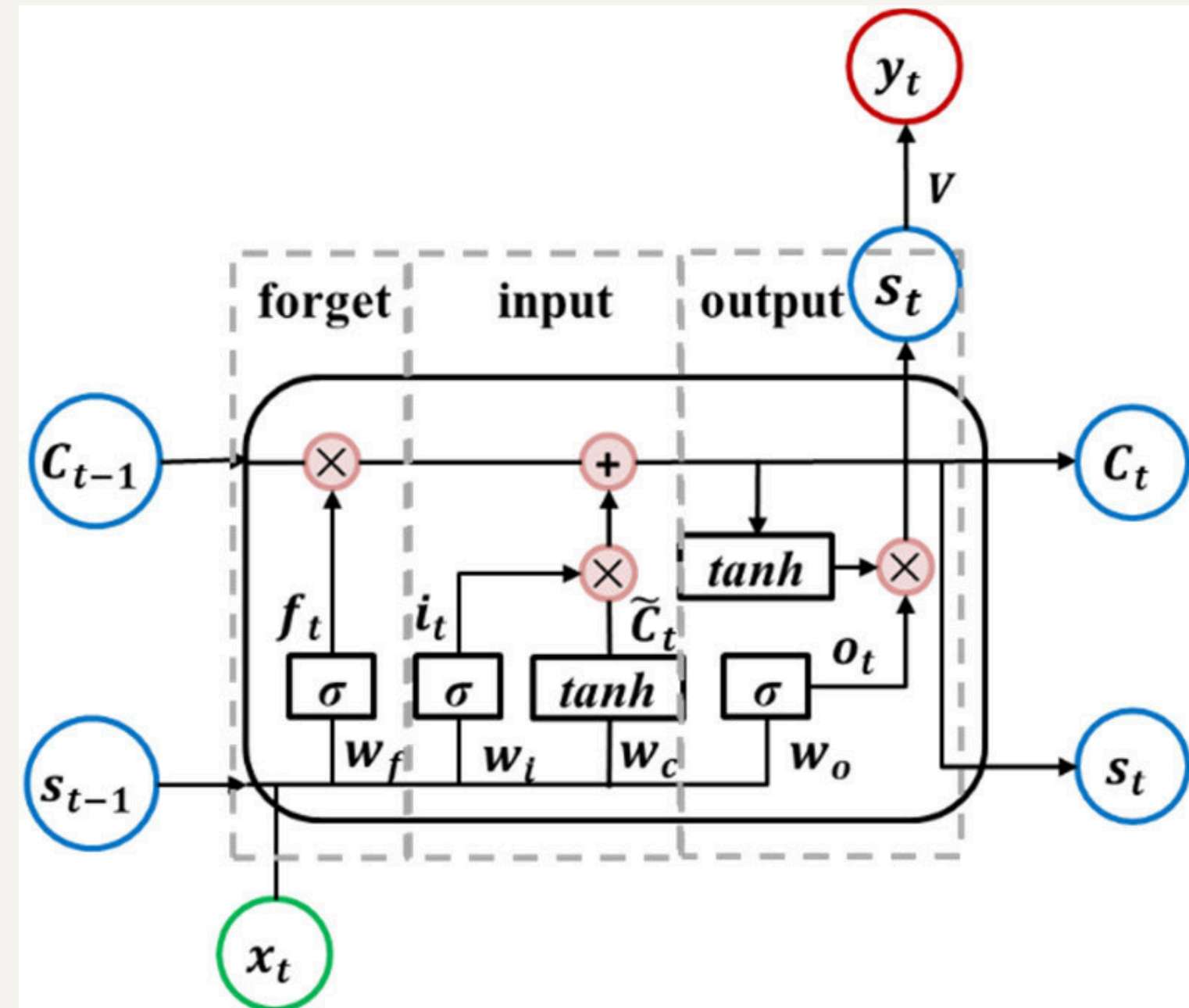
Gated Recurrent Unit (GRU)

AI models

Deep Learning (Recurrent Neural Network)

Processes sequential data using memory units to learn patterns over time

Good for time-series and sequential data



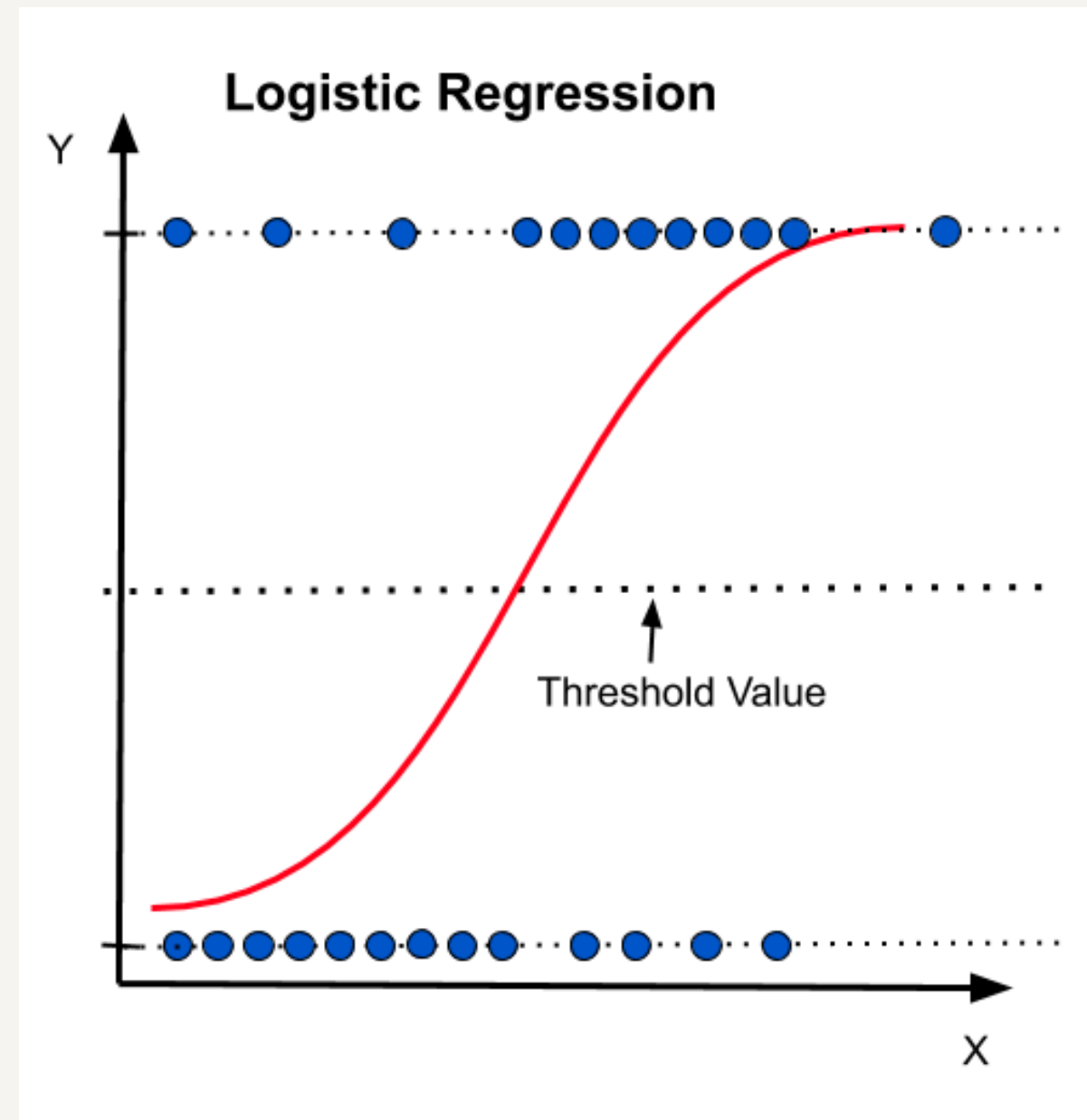
Logistic Regression

AI models

Linear Model

Uses a logistic function to model probabilities for binary or multi-class classification

Assumes linear decision boundary, less effective on complex data



Model Performance Analysis

MMASH Dataset

ACTIVITIES

Sitting
Walking
Lying down
Eating
Small screen usage
Large screen usage
Caffeinated drink consumption
Smoking
Alcohol assumption

SENSORS

Accelerometers | Gyroscopes | Magnetometers | Heart Rate

MODEL PERFORMANCE

XGBoost and LightGBM:

Higher accuracy

Random Forest and Gradient Boosting:

Best performance for generalized activities with large support value

KNN:

Low precision and recall

Model Performance Analysis

ScientISST Dataset

ACTIVITIES

Sitting
Walking
Running
Lift a chair
Greetings
Gesticulate
Jump

SENSORS

Accelerometers | GYR | Temperature | ECG | EDA | PPG

MODEL PERFORMANCE

KNN and Random Forest:

Top performance models, nearly perfect performance

CNN and GRU:

CNN excelled in spacial feature extraction, and GRU in transitions

Challenges in Gesture Recognition:

Low All models struggled with nuance activities like 'greetings' and 'jump'

Model Performance Analysis

PAMAP2 Dataset

ACTIVITIES

Lying
Walking
Running
sitting
standing
cycling
nordic walking
whatching tv
computer work
car driving
ascending stairs
descending stairs
vacuum cleaning
playing soccer
rope jumping

SENSORS

Accelerometers | Gyroscope | Heart Rate

MODEL PERFORMANCE

Random Forest and XGBoost:

Nearly Near perfect precision, recall and F1-score for diverse activities

LightGBM:

Optimized for speed and memory efficiency, ideal for real-time activity tracking

KNN and Logistic Regression:

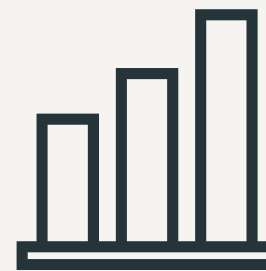
Low KNN performed well on frequent tasks but struggled with transitions, while Logistic Regression had issues differentiating nuance activities.

Discussion and Future Work



DATASET LIMITATION

Current datasets lack elderly representation, diverse activities, and real-world variability



MODEL STRENGTHS AND WEAKNESSES

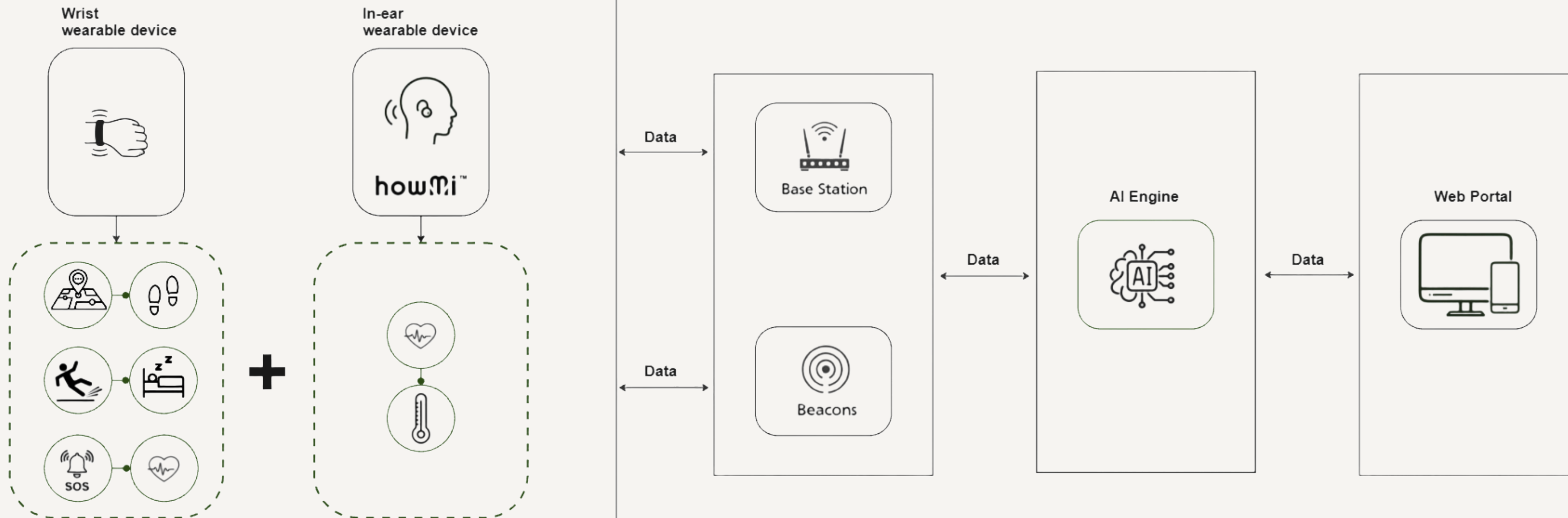
RF, XGBoost and LightGBM excel in general activity tracking but struggle with nuanced tasks



NEED FOR IMPROVED DATA COLLECTION

Longer-duration recordings in real-world environments are essential for robust AI models

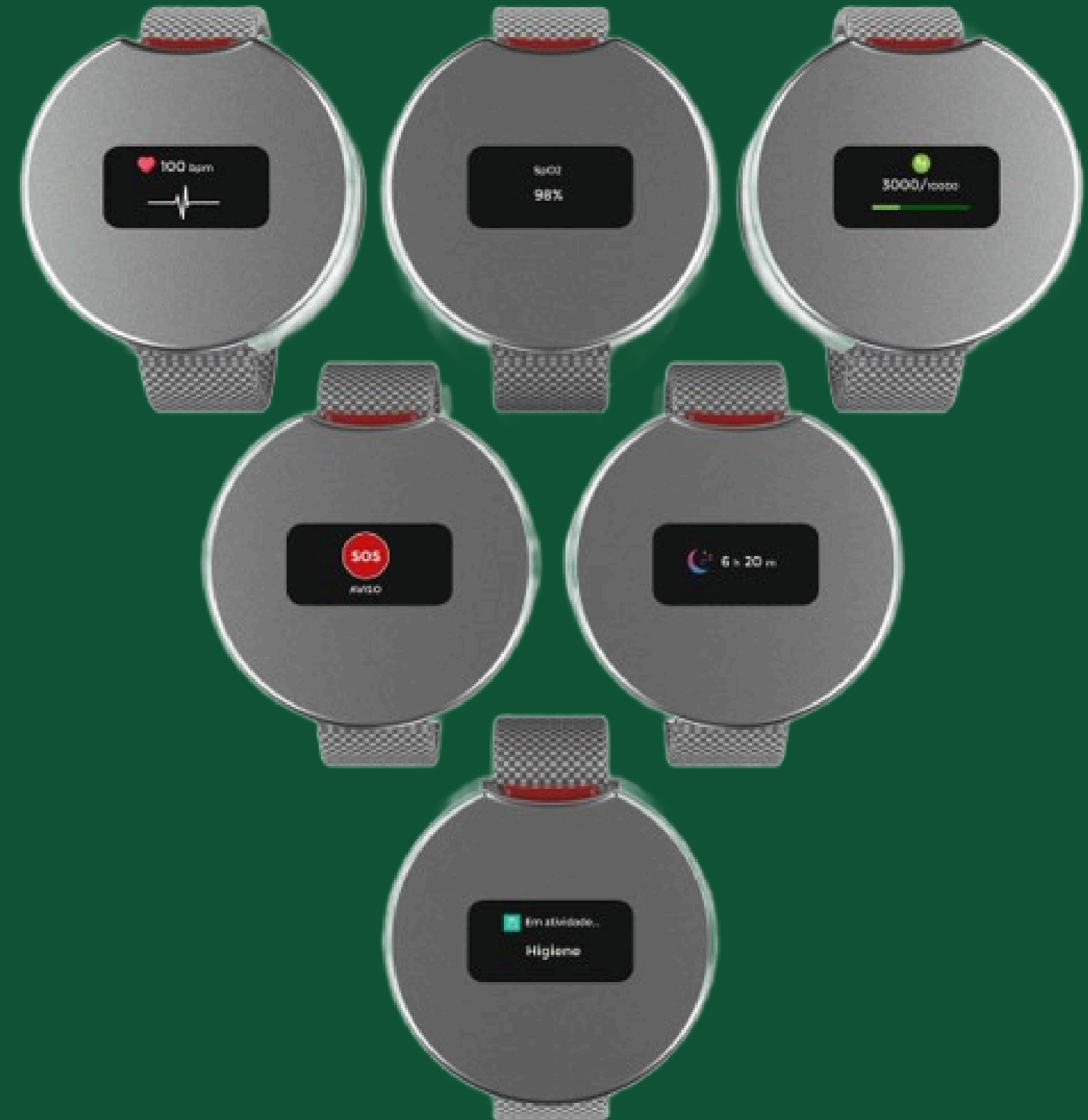
Monitoring System



Wearable Device

Inertial sensors for sensory fusion and subsequent analysis of **daily activity patterns**.

- Activity index, number of steps and distance covered;
- Analysis of sleep patterns;
- Fall detection;
- Heart rate monitoring;
- SOS warnings;



Wearable Device



SOS button for quick alert

Audible Alert Warnings

Multisensory analysis for **device removal detection** and caregiver notification

Device autonomy management (+24h)

Defining a pattern of daily activities and behaviors through the use of AI

Powered Wearable Technologies for Dementia Care: Evaluating Activity Recognition Models and Dataset Challenges

THANK YOU !



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